

Simple Linear Regression

Supervised ML $\rightarrow$ Regression

Dataset I/P features

$\uparrow x$
Weight

74

80

75

-

y

$\uparrow$ O/P or
Height
dependant
feature

170cm

180cm

175.5cm

-

New
weight

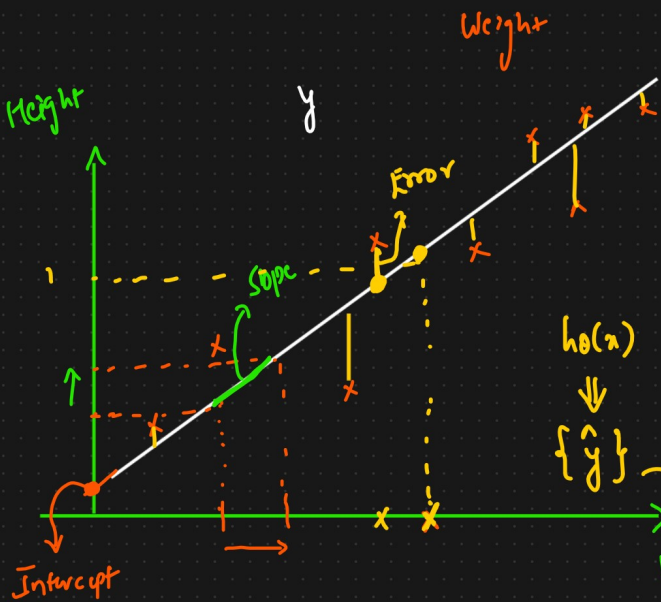
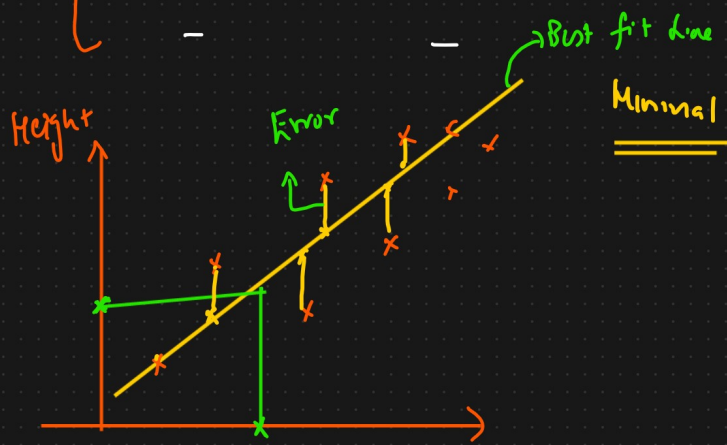
TRAIN



Model

Simple Linear
Regression

Height



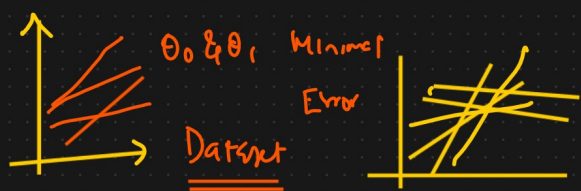
$$h_0(x) = \theta_0 + \theta_1 x$$

θ_0 = Intercept

θ_1 = slope or coefficient

if $x = 0$

$$h_0(x) = \theta_0$$



$$y = mx + c$$

$$y = \beta_0 + \beta_1 x$$

$$h_0(x) = \theta_0 + \theta_1 x$$

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

$$\text{Error} (y - \hat{y})$$

Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (\overset{\text{predicted}}{\uparrow} h_0(x)^{(i)} - \overset{\text{True O/P}}{\uparrow} y^{(i)})^2 \Rightarrow \text{Mean Squared Error}$$

Error

Final Aim what we need to solve

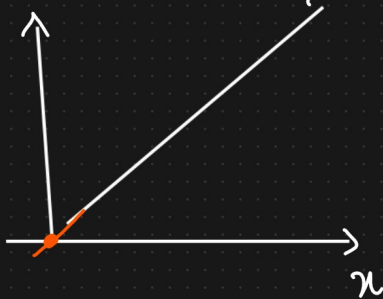
Minimize $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x)^i - y^{(i)})^2$ ↓↓↓

θ_0, θ_1

① $h_0(x) = \theta_0 + \theta_1 x$

$\theta_0 = 0$

$h_0(x) = \theta_1 x$



$\theta_0 = 0$

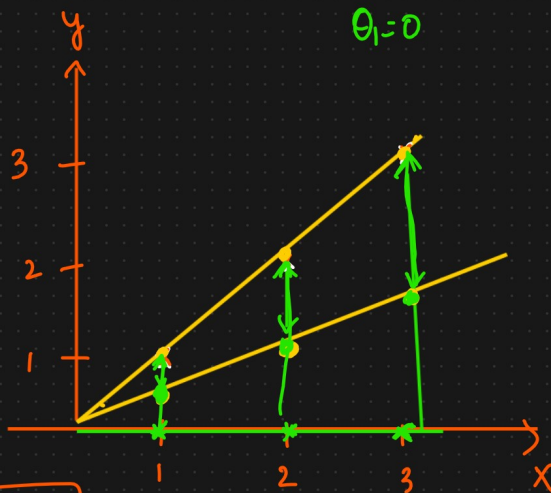
$h_0(x) = \theta_1 x$

Let $\theta_1 = 1$ {slope}

$h_0(x) = 1$ if $x = 1$

$h_0(x) = 2$ if $x = 2$

$h_0(x) = 3$ if $x = 3$



$\theta_1 = 1$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})^2$$

$$= \frac{1}{2 \times 3} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

DATASET

x	y
1	1
2	2
3	3

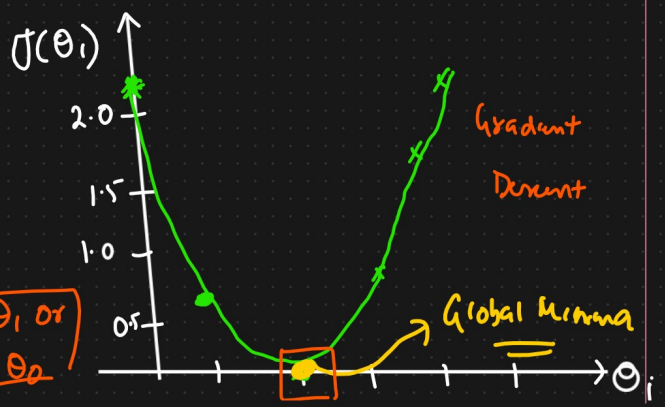
$h_0(x) = \theta_1 x$

Let $\theta_1 = 0.5$

$h_0(x) = 0.5$ if $x = 1$

$h_0(x) = 1$ if $x = 2$

$h_0(x) = 1.5$ if $x = 3$



θ_1 or θ_2

$$\underline{J(\theta_1)} = 0 \leftarrow$$

$$\theta_1 = 0.5$$

Error has been $0.5 \quad 1 \quad 1.5 \quad 2.0 \quad 2.5$
minimized

$$J(\theta_1) = \frac{1}{2 \times 3} \left[(0.5-1)^2 + (1-2)^2 + (1.5-3)^2 \right]$$

$$J(\theta_1) = \approx \underline{0.58}$$

if $\theta_1 = 0$

$$J(\theta_1) = \frac{1}{2 \times 3} \left[(0-1)^2 + (0-2)^2 + (0-3)^2 \right]$$

$$J(\theta_1) \approx \underline{2.3}$$

Convergence Algorithm { Optimize the changes of θ_1 values }

Repeat until convergence

{

$$\left\{ \theta_j := \theta_j - \alpha \left[\frac{\partial J(\theta_j)}{\partial \theta_j} \right] \rightarrow -ve \right.$$

}

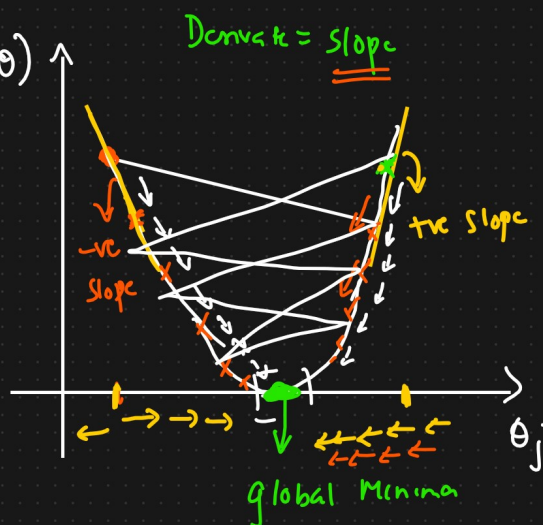
$$\theta_j = \theta_j - \alpha (+ve)$$

$$= \theta_j - (+ve)$$

α = Learning Rate

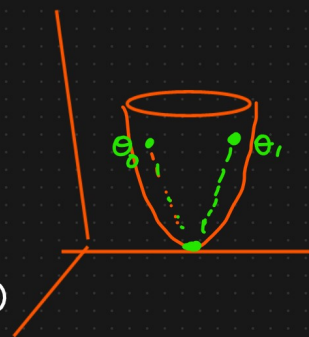
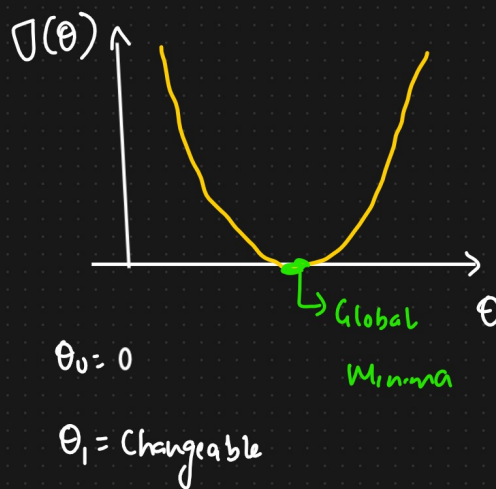
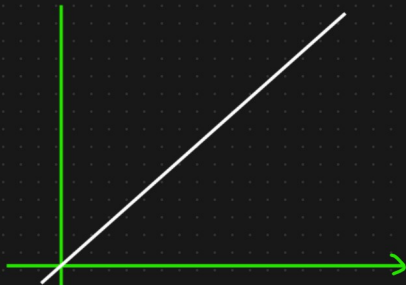
$$\left\{ \begin{array}{l} \theta_j = \theta_j - \alpha (-ve) \\ \theta_j = \theta_j + (+ve) \end{array} \right\}$$

$$\underline{\underline{\alpha = 0.001}} \leftarrow$$



Final Conclusion

GRADIENT DESCENT



Convergence Algorithm

repeat until convergence

{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$j = 0$ and 1

$$\frac{\partial}{\partial x} x^2 = 2x$$
$$\frac{\partial}{\partial x} x^h = h x^{h-1}$$

$$\rightarrow \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

if

$$h_{\theta}(x) = \theta_0 + \theta_1 x \rightarrow 0$$

$$j=0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_0} \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \right]$$
$$= \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \times 1$$

$$j=1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_1} \left[\frac{1}{2m} \sum_{i=1}^m (\theta_0 + \theta_1 x^{(i)} - y^{(i)})^2 \right]$$

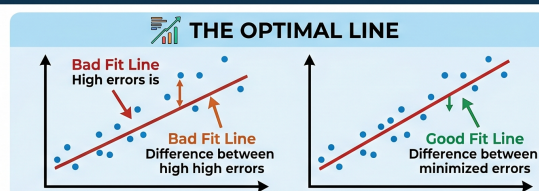
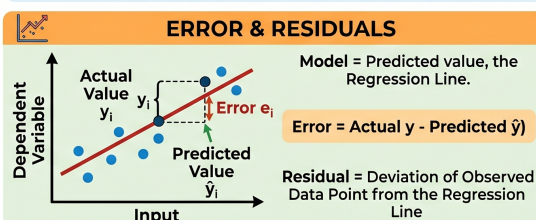
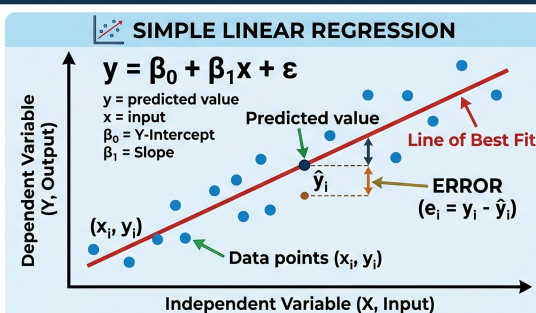
$$= \frac{1}{m} \sum_{i=1}^m ((\theta_0 + \theta_1 x^{(i)}) - y^{(i)}) (x^{(i)})$$

$$\frac{\partial}{\partial \theta_1} [\theta_0 + \theta_1 x] \Rightarrow x =$$

Repeat until Convergence

$$\left\{ \begin{aligned} \theta_0 &:= \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \\ \theta_1 &:= \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)} \end{aligned} \right\}$$

UNDERSTANDING SIMPLE LINEAR REGRESSION MODEL

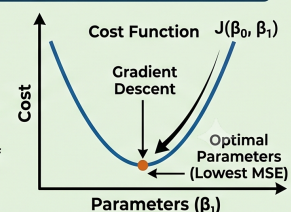


COST FUNCTION (MEAN SQUARED ERROR - MSE)

$$MSE = J(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (e_i)^2$$

THE GOAL:
MINIMIZE
THE COST!

MSE calculates the average of squared errors to quantify the model performance, that it calculates



$$MSE = (1/m) \times \sum \text{Error}^2$$

$$\text{Cost Function} = (1/2m) \times \sum \text{Error}^2$$